

Elia Gatti

Master's Student in Computer Science | HPC & GPU Software Engineer

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Work Experience

Bitmovin, Software Engineer Intern

June 2026 to Sept 2026

- Shipped KAIROS an **AI video highlight detection and segmentation** for Sports, News and Podcasts with VOD plus Live analysis, public API, MCP and one click highlight reels.
- Cut transcription latency by **20%** by tuning parallelism and slimming workers and live playback by measurement, improving cost and reliability.
- Drove **33k views to 130 signups in two weeks** by launching Google Ads campaign with signup tracking and building feedback loop to Slack plus admin traction charts.
- Improved **pipeline reliability** by building job orchestration with startup fixes and live observability, reducing failed runs and manual ops work.

Dedagroup, Internship Software Developer

May 2024 to Sept 2024

- Migrated the 'TEN' treasury application from legacy Windows to Linux servers, **halving VM resource requirements**.
- Developed reactive **HTMX frontend features** and maintained the Java/Spring backend.

Education

Expected 2026
Graduated 2024

Master's Degree in Computer Science
Bachelor's Degree in Computer Science

University of Trento
University of Trento

Projects

Sparse Matrix Vector Multiplication (SpMV) Optimization [↗](#)

- Developed a **hybrid adaptive CUDA kernel** for NVIDIA A30.
- Profiled execution time and GFLOPS, focusing on memory access coalescing and occupancy tuning.

Parallel Minimum Spanning Tree (CPU) [↗](#)

- Implemented parallel MST algorithms (Borůvka / Kruskal) using OpenMP and MPI.
- Analyzed speedup and scalability on multicore CPU clusters.

Distributed Key Value Storage System [↗](#)

- Built with **Akka actors**: consistent hashing, data replication, fault tolerance.
- Handled concurrent read and write requests and node failures without client disruption.

Guardian: Privacy First Eldercare [↗](#)

EuroTech HK Hackathon, Munich, June 2026

- Built zero config elderly monitoring with mmWave radar plus on device Gemma 4 LLM: no cameras, no wearables, no cloud.
- Cross border eldercare: behavioral drift tracking with real time crisis alerting.

DWT SVD Watermarking (1st Place) [↗](#)

CTM Challenge, Trento, 2025

- Engineered SVD based watermark embedding within DWT blocks, robust to compression, noise and transforms.
- Built automated attack suite (ROC validation).

Skills

Languages

Frameworks & Tools

Concepts

C, C++, CUDA, Java, Python

FastAPI, Docker, Git, Linux, OpenMP, MPI

GPU programming (memory hierarchy, kernel optimization), parallel algorithms, distributed systems, performance benchmarking